# LSS 2026 Workshop: Data Visualization

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## 0.1 Welcome!

#### 0.1.0.1 Who am I?

- My name is Grace
- I'm a Biostatistician and Assistant Professor of Teaching (Dept. of Statistics)

#### 0.1.0.2 Who is this workshop for?

- If you're new(ish) to R or relatively experienced, you're in the right place!
- This workshop will go at *your* pace - just keep in mind we only have an hour together

## 0.2 Goals of this Workshop

By the end of this workshop, attendees should be able to:

- Interact with data in R more confidently
- Create plots using `ggplot2`

## 0.3 How to Interact with R in This Workshop

- These slides are built with Quarto + WebR, meaning you can code in them without downloading any software!
- Your code *should* stay on the web address, but I recommend saving a PDF by [right clicking] -> [Print…] -> [Save/Export to PDF] :)
- For your own research/projects, I highly recommend checking out how to use RStudio! Here's a video on how to use RStudio.

## 0.4 Data Visualization: Why Do We Use It?

- Tables are hard to digest.
- Visualizations can convey more complex ideas faster
- Visualizations can help us answer descriptive and exploratory questions
  - They do not answer predictive, inferential, or causal questions

## 0.5 Great Data Visualizations

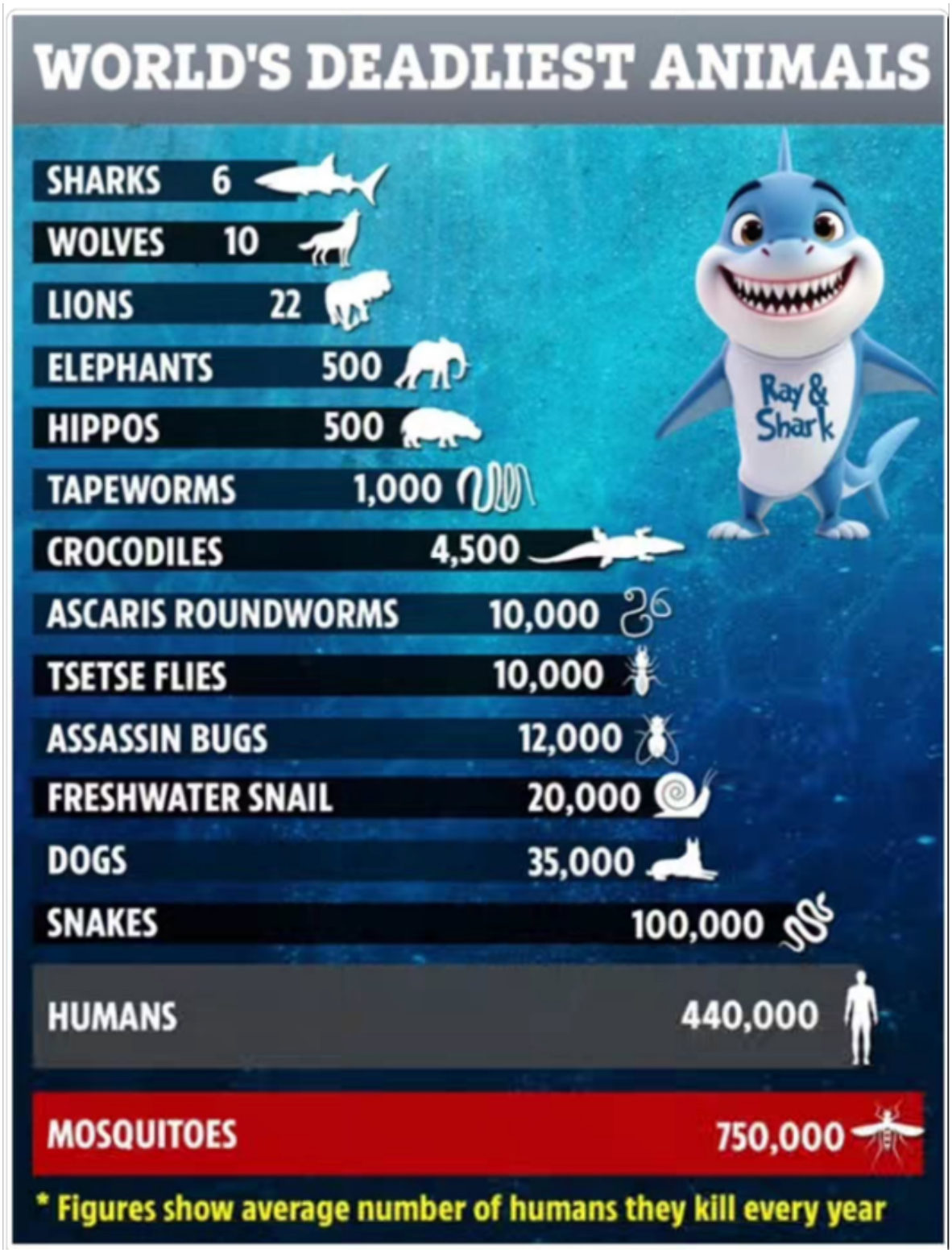
"A good visualization will clearly answer your question without distraction; a great visualization will suggest even what the question was itself without additional explanation." - *A First Introduction to Data Science*

How do we accomplish this? Keep this **SASS**-y

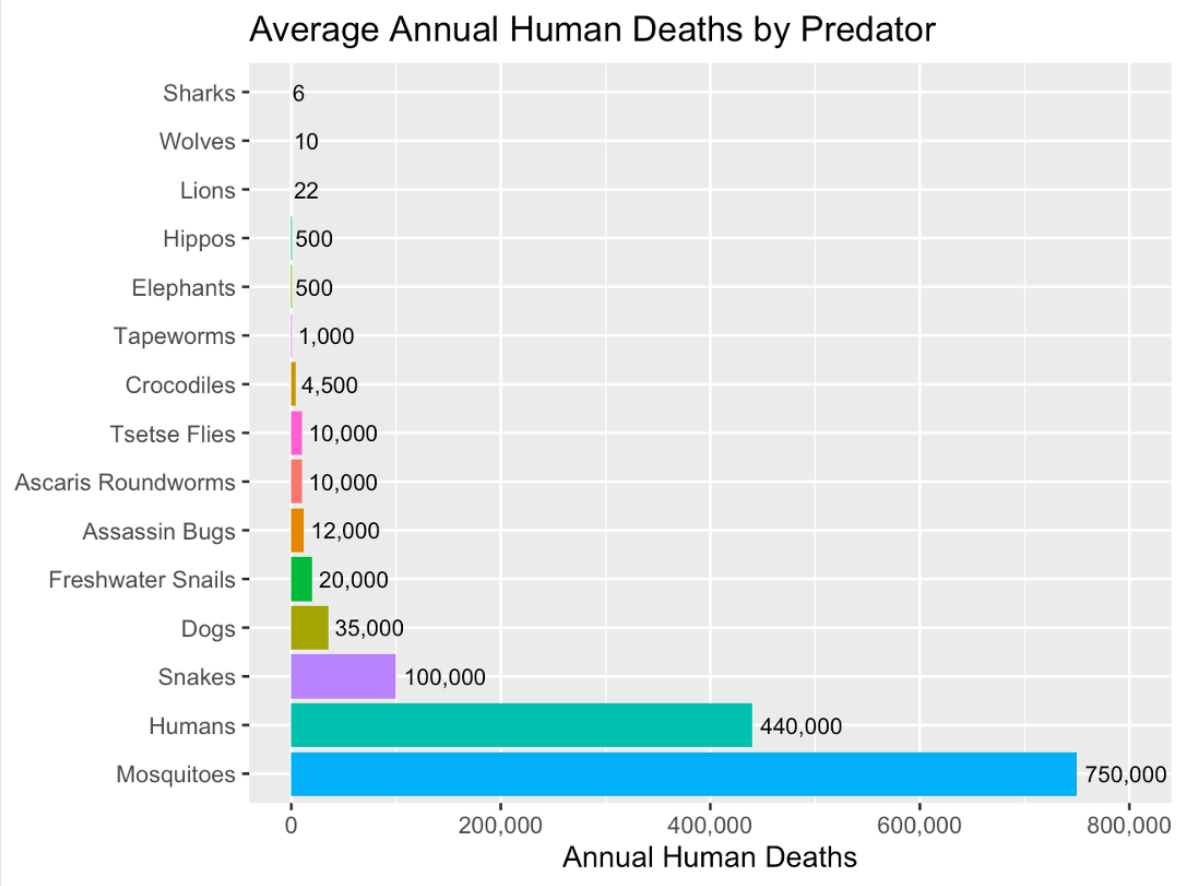
- Simple (plot is as simple as possible, minimizing distractions)
- Accessible (colourblind-friendly pallettes are used, text is human-readable)
- Specific (the purpose of the plot is clear, and explores a specific research question)
- Scaled (small differences are not blown up, proportionality is maintained)

## 0.6 Data Visualizations

Which principles of good data visualization are met/violated here?


WORLD'S DEADLIEST ANIMALS
SHARKS 6
WOLVES 10
LIONS 22
ELEPHANTS 500
HIPPOS 500
TAPEWORMS 1,000
CROCODILES 4,500
ASCARIS ROUNDWORMS 10,000
TSETSE FLIES 10,000
ASSASSIN BUGS 12,000
FRESHWATER SNAIL 20,000
DOGS 35,000
SNAKES 100,000
HUMANS 440,000
MOSQUITOES 750,000
* Figures show average number of humans they kill every year
Ray & Shark

## 0.7 Data Visualizations




(Numbers based off previous image which is… questionably sourced :-) )

## 0.8 Making Nice Plots in R

A crowd favourite package for plotting in R is `ggplot2` (which has the `ggplot()` function).

There are three key aspects of plots in `ggplot2`:

1. **aesthetic mappings:** relates dataframe columns to visual properties
2. **geometric objects:** disctates how to display those visual properties (type of plot, for example)
3. **scales:** transforms variables, sets limits

We add these layers one by one using `+`

> **Important Note**
>
> Building plots is an iterative procedure. Try things, make mistakes, and refine!

# 1 Demo

## 1.1 Today's Demo: Penguins!

We will explore the `penguins` data set from the `palmerpenguins` package in R.

```
# install the required package (first time only)
# install.packages(c("palmerpenguins", "ggplot2", "forcats"))
#I've already installed it for you :)

# load in the required package to get access to the penguins dataset
library(palmerpenguins)
library(ggplot2)
library(forcats)

# look at the first few rows of the data with head()
head(penguins)
```

## 1.2 Research Questions

An interesting research question may be how some of these measurements within a penguin are related. Let's investigate the following:

**Descriptive Visualizations**

- What species of penguins are in my data set? How many of each sex?
- What does the distribution of bill length measurements look like?

**Exploratory Visualizations**

- Do penguins with longer bills also tend to have wider bills?
- Do certain species of penguin tend to have larger bills?

## 1.3 Choosing a Data Visualization

Typically a data visualization is 2-Dimensional (think of drawing a plot on a piece of paper). We can easily visualize the relationship between two variables.

**What plot should I use?**

- **Scatterplots**: used to visualize two quantitative (numeric) variables
- **Line plots**: used to visualize trends with respect to an independent quantity (like time)
- **Bar plots**: used to visualize the comparison of amounts (categorical variables). Can be stacked or grouped to show the relationships across another categorical variable.
- **Box plot** or **histograms**: used to visualize distributions, perhaps across groups.

## 1.4 Choosing a Data Visualization: Counting Penguin Species

Let's visualize the types of penguins (species and sex) in our study.

If we wanted to show the **counts** of each penguin species, which plot should we use?

. . .

A bar plot! We can plot this using `geom_bar()` in `ggplot`.

## 1.5 Bar Plot (`geom_bar()`)

Let's visualize how many penguins of each species we have in our data set

```
ggplot(penguins, aes(x = fct_infreq(species), fill = species)) +
     geom_bar() +
     ylab("Number of Penguins") +
     xlab("Species") +
     ggtitle("Number of Penguins by Species") +
     coord_flip() +
     geom_text(stat = "count",
        aes(label = after_stat(count)),
        hjust = -0.1)
```

## 1.6 Bar Plot (`geom_bar()`): Elevated!

Let's get the counts by sex, too!

```
ggplot(penguins, aes(x = fct_infreq(species), fill = species)) +
     geom_bar() +
     ylab("Number of Penguins") +
     xlab("Species") +
     ggtitle("Number of Penguins by Species") +
     coord_flip()
```

```
ggplot(penguins, aes(x = fct_infreq(species), fill = sex)) +
     geom_bar(position = position_dodge(preserve = "single")) +
     ylab("Number of Penguins") +
     xlab("Species") +
     ggtitle("Number of Penguins by Species") +
     labs(fill = "Sex") +
     geom_text(stat = "count",
               aes(label = after_stat(count)),
               position = position_dodge(width = 0.9, preserve = "single"),
               hjust = -0.1) +
     coord_flip()
```

## 1.7 Choosing a Data Visualization

Let's describe the distribution of the bill lengths in our data set. What visualization should we use?

. . .

Histogram! (Or… a box plot)

## 1.8 Histogram (`geom_hist()`)

```
library(ggplot2)
library(palmerpenguins)
```

```
ggplot(penguins, aes(x = bill_length_mm)) +
  geom_histogram(binwidth = 1, fill = "steelblue", color = "white") +
  labs(
    title = "Distribution of Penguin Bill Lengths",
    x = "Bill Length (mm)",
    y = "Frequency"
  )
```

### 1.9 Choosing a Data Visualization

Let's visualize bill depth and bill width, which are measured in millimetres.

What visualization should we use?

. . .

Scatterplot! We plot this using the `geom_point()` object within `ggplot`.

### 1.10 Scatterplot (`geom_point()`)

```
# Create a basic scatterplot using ggplot
```

```
ggplot(penguins, aes(x = bill_length_mm, y = bill_depth_mm)) + # call the
  # ggplot function on the penguins data set, assign the bill length
  # to the x axis, bill depth to the y axis
  geom_point() + # plot a scatterplot
  xlab("Bill Length (mm)") + # update the x axis label name
  ylab("Bill Depth (mm)") + # update the y axis label name
  ggtitle("Relationship between penguin bill lengths and depths") # add a title
```

### 1.11 Scatterplots: Grouping by a Categorical Variable

What about looking at this relationship between species? Do certain species of penguin tend to have larger bills?

. . .

We can group by colour/shape to see if there are trends within/between penguin species!

```r
# Elevate the basic scatterplot to show species

ggplot(penguins, aes(x = bill_length_mm, y = bill_depth_mm)) +
  geom_point() + # plot a scatterplot
  xlab("Bill Length (mm)") + # update the x axis label name
  ylab("Bill Depth (mm)") + # update the y axis label name
  ggtitle("Relationship between penguin bill lengths and depths")
```

```r
ggplot(penguins, aes(x = bill_length_mm, y = bill_depth_mm,
                     col = species, shape = species)) + # call the
  # ggplot function on the penguins data set, assign the bill length
  # to the x axis, bill depth to the y axis. let species dictate the shape
  # and color of the point
  geom_point() + # plot a scatterplot
  xlab("Bill Length (mm)") + # update the x axis label name
  ylab("Bill Depth (mm)") + # update the y axis label name
  ggtitle("Relationship between penguin bill lengths and depths") + # add a title
  # update the title of the legend
 labs(color = "Species", shape = "Species")
```

### 1.12 Try it yourself:

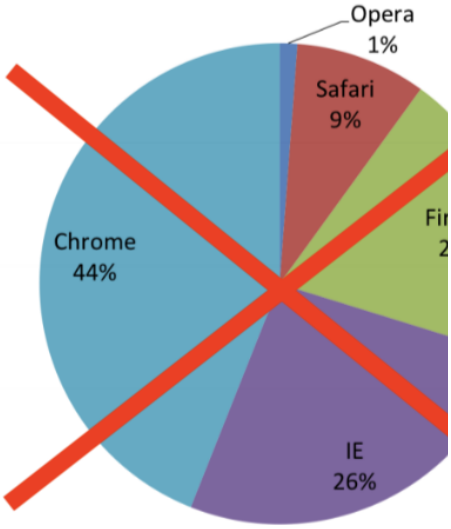
Are body mass and flipper lengths in penguins related? Does this vary by sex?

```r
# Show this relationship!
```

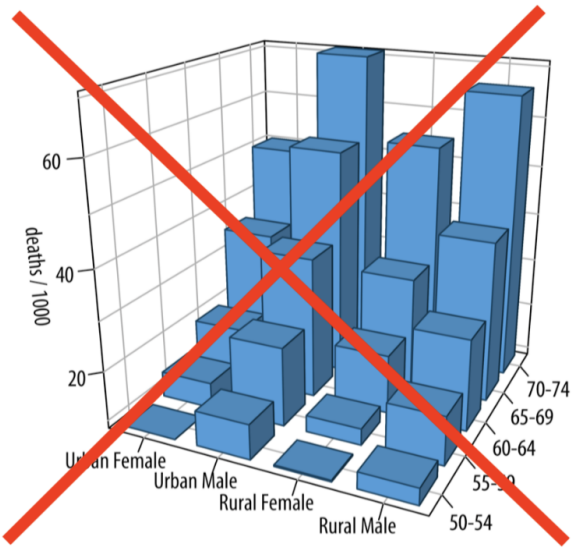
```r
ggplot(penguins, aes(x = flipper_length_mm, y = body_mass_g, col = sex)) + geom_point() +
  xlab("Flipper Length (mm)") + # update the x axis label name
  ylab("Body Mass (g)") + # update the y axis label name
  ggtitle("Relationship between penguin flipper length and body mass") +
  # update the title of the legend
 labs(color = "Sex", shape = "Sex")
```

### 1.13 Other Data Visualization Tips

- Avoid pie charts.
- Avoid 3D visualizations
- ….. Absolutely avoid 3D pie charts




(a) Ugly Pie Chart




(a) Disgusting 3D Plot

## 1.14 Looking for more?

**Before you click any links, save your work as a pdf!**

[Right Click] -> [...Print] -> [Save/Export as PDF] (or right click to open links in a new tab)

- STAT545 Coursenotes on Data Visualization: https://ubc-stat.github.io/stat545/webpages/lectures_i/lec4_datavis.html (I teach STAT 545 A/B: Exploratory Data Analysis every fall, which is a data science-y course for non-statisticians)
- Jenny Bryan's "Dos and Don'ts of Making Effective Graphs": https://stat545.com/effective-graphs.html
- R4DataScience Notes on Data Viz: https://r4ds.had.co.nz/data-visualisation.html
- LLMs like Claude, which are quite good at creating data visualizations.
  - NEVER upload private data to an LLM!
- My YouTube channel @proftompkins

## 1.15 Thank you!